TRANSPORTATION MODELING

OVERVIEW

Using only designated materials and following required specifications, participants apply leadership and 21st century skills in the research, design, and production of a scale model of a vehicle that fits the annual design problem, which will be posted on the TSA website under *Themes & Problems*. The entry must take appearance and realism into consideration.

ELIGIBILITY

One (1) individual per chapter may participate.

TIME LIMITS

Semifinalists participate in an on-site interview that lasts up to five (5) minutes.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Participants produce a scale model of a vehicle and supporting display to contain it, focusing on the year's current theme, while observing the outlined regulations.

D. Participants prepare the documentation portfolio and display according to the regulations for this event.

PRELIMINARY ROUND

A. Participants check in the following at the time and place stated in the conference program:

   1. the scale model
   2. the display
   3. the documentation portfolio

B. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:

   1. Judges score the Model and Display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
   2. Judges score the Documentation Portfolio criteria of those top twenty-four contestants to determine the top twelve (12) semifinalist teams.

C. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

A. Participants report at the time and place stated in the conference program to sign up for an interview time.

B. Judges ask questions pertaining to the research, production of the model, and the design process.

C. The top ten (10) finalists are announced during the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. Entries must include a scale model, a display, and documentatio portfolio.

B. Model, display and documentation must meet the following specifications:

Model

   1. Scale model must accurately reflect the annual design problem.
   2. Model must be designed and produced as original work by the student during the current school year.
   3. Model may be made from wood, urethane modeling foam, or it may be 3D-printed.

4. Using commercially produced (store-bought) model vehicle body parts (including hoods, fenders, wings, propellers, frames, etc.) is prohibited.
5. It is permissible to use pre-manufactured parts such as body strengtheners, tires and wheels, plastic canopy, exhausts, mirrors, head and tail lights, windshields, and antennae.
    a. These parts may be attached to or enclosed within the vehicle and may be constructed from materials other than wood, excluding glass or liquids. These parts must be fastened securely.
    b. It is also permissible to use CNC production and 3D printers in the production of the parts of this model.
6. Finished vehicle must fit inside the display space of 16" x 16" x 16".
7. Themed vehicle model must have an actual length that measures at least six inches (6").
8. Designer must choose a scale for the vehicle so that it meets regulations and must be specified in the portfolio.
9. Wheels: Dimensions should be consistent with the scale of the body.

Display

1. Model must be presented for evaluation on a display not to exceed 16" tall x 16" deep x 16" long (including the model).
2. Portfolio is not considered part of the display but is placed with it at its side.
3. No electrical access will be provided by TSA for displays.
4. Use of dry cell batteries is permissible, but they must be contained within the stated display space.

Documentation Portfolio

1. Documentation materials (comprising a "portfolio") are required and must be secured in a clear front report cover.
2. In addition to the 11" x 17" pages noted below, the report cover must include the following single-sided, 8½" x 11" pages, in this order:
    a. Title page with the event title, the conference city and state, and the year; a picture of the vehicle may be included as well; one (1) page
    b. Table of contents; pages as needed
    c. Description of designer's vehicle, making note of the scale used, inspiration for the choice and design of the vehicle, research about the history and evolution of the original vehicle, and design elements that set the vehicle apart from others (e.g. fuel used, unique features); one (1) page
    d. Photo examples of current or past vehicles that are similar to the current year's theme or that inspired the entry; one (1) page
    e. Concept drawings/detailed sketches or 3D CAD modeling – must be on 11" x 17" paper; maximum of two (2) pages
    f. Photos of the clay, foam, wax, or 3D-printed mock-up; one (1) page
    g. Final technical illustrations (orthographic) must be on 11" x 17" paper; maximum of two (2) pages
    h. Photos of the production of the model; one (1) page
    i. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages
3. Documentation for this event must not include the name of the student name, chapter, or state.
4. All ideas, text, or images from sources other than the designer must be cited.
5. Cited works should be in MLA format.
6. Pages that are 11" x 17" in size should be folded to fit in the notebook.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. Model and display

Tier 2

B. Documentation portfolio

SEMIFINAL ROUND

A. Interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Automotive designer
- Automotive engineer
- Digital modeling technician
- Industrial designer
- Industrial engineer

Participant/Team ID# _____

TRANSPORTATION MODELING

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ TIER 1 – Model and Display
☐ TIER 2 – Documentation portfolio
☐ ENTRY NOT EVALUATED

TIER 1 – MODEL AND DISPLAY (50 points)				
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Production Quality (X1)	Model exhibits poor production quality; surface is rough; there is little or no attention to detail.	There is some evidence of proper production techniques; model appearance is adequate.	Model demonstrates excellent production techniques with obvious effort and attention to detail.	
Paint and Finish (X1)	Surface imperfections are evident; model is sticky, and/or the painting quality is low.	Quality of the painted surface is acceptable, with some imperfections visible.	Painted surface is exceptional, with little or no visible imperfections.	
Appropriate to Designated Problem (X1)	Model does not relate to the stated annual design theme.	Model generally relates to the stated annual design theme.	Model effectively represents and portrays the stated annual design theme.	
Details (X1)	There is a very weak and limited attempt to include identifying characteristics and/or additional parts to help create a realistic appearance.	Model includes some identifying characteristics and/or additional parts that give it a sense of realism.	Model displays exemplary effort to include identifying characteristics and/or additional parts that give it a realistic appearance.	
Display (X1)	Quality of the display is poor, and/or it exceeds the size requirements.	Display is adequately created and meets the size specifications.	Display is exemplary, includes eye-catching details, and meets the size specifications.	
			TIER 1 – MODEL AND DISPLAY SUBTOTAL (50 points)	

Rules violations (a deduction of 20% of the total possible points in the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIER 1 – SUBTOTAL (50 points)

TIER 2 – DOCUMENTATION PORTFOLIO 80 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Vehicle Description (X1)	Description is inadequate; research references are lacking; scale is incomplete.	Description is adequate, research is evident with some documentation, and the scale is stated and accurate.	An excellent description is included, with necessary research referenced to support the model solution; scale is stated and accurate.	
Concept Drawings, Detailed Sketches, or 3D CAD Modeling (X1)	Drawings are not to scale, and/or the the quality is poor, and/or there are missing parts and dimensions; drawings are not on 11" x 17" paper.	Drawings are acceptable, true to scale, and representative of the vehicle, with some details/ dimensions included; drawings are produced on 11" x 17" paper.	Drawings are accurate and complete; they include all necessary details/dimensions and are drawn on 11" x 17" paper.	
Photo Examples of Current/Past Similar Vehicles (X1)	There is only one (1) photo example of current or past similar vehicles.	There are two or three (2-3) photo examples of current or past similar vehicles.	There are a number of photo examples of current or past similar vehicles, showing that in-depth research was done.	
Photos of Clay, Foam, Wax, or 3D-Printed Model (X1)	There is only one (1) photograph of the clay/foam or wax model included.	Two or three (2-3) photographs of the clay/foam or wax model are included, but more are needed to adequately document the model.	There are a number of photographs included that effectively document the preliminary clay/foam/wax model.	
Final Technical Illustrations (orthographic plans) (X1)	Orthographic plans are poorly executed, and/or the plans are not on 11" x 17" paper.	Adequate orthographic plans are included; plans are on 11" x 17" paper.	Complete orthographic plans are included; they are of excellent quality on 11" x 17" paper.	
Photos of Production of the Model (X1)	Only one (1) photograph of the model production is included.	Two or three (2-3) photographs of the model production are included, but they are not enough to provide full documentation.	Photographs included fully and effectively document and describe the model production process.	
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
			TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (80 points)	

Rules violations (a deduction of 20% of the total possible points in the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (130 points)

SEMIFINAL INTERVIEW (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21st century skills are clearly evident.	
Knowledge (X1)	Participant seems to have little understanding of the development and concepts in the project; vague interview answers are provided.	Participants have a generalized understanding of the dvelopment and concepts discussed and answer questions adequately.	Evidence is clear that the participant has a thorough understanding of the development and concepts discussed; questions are answered thoroughly.	
Organization (X1)	Participant is unprepared and unorganized for the interview, with an illogical explanation of the project.	Participant is prepared for the interview and is somewhat organized in his/her explanation to judges; answers are, for the most part, logical and/or clear.	Participants responses are concise and logical, with a clear explanation of the development of the project.	
			SEMIFINAL INTERVIEW SUBTOTAL (30 points)	

Rules violations (a deduction of 20% of the total possible points in the semifinalist sections above) must be initialed by the evaluator, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
SEMIFINAL SUBTOTAL (30 points)	
To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (160 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

TRANSPORTATION MODELING
EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Assistants, two (2)
C. Judges:
   1. Preliminary round, two (2) or more
   2. Semifinal round, two (2) or more

MATERIALS

A. Coordinator's packet containing:
   1. Event guidelines, one (1) copy for the coordinator and for each judge
   2. TSA Event Coordinator Report
   3. Stick-on labels for entries
B. Stopwatch
C. Measuring tape
D. Interview questions

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough personnel have been scheduled.
D. Inspect the area(s) in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Check in the entries at the time and place stated in the conference program.
B. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.
C. Participants check in:
   1. the scale model
   2. the documentation portfolio
   3. the display
D. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.
E. In order to compete, participants must be on the entry list or must have CRC approval.
F. Each entry must include the identification number in the upper right-hand corner of the entry.
G. Instruct the participants to position the displays for viewing.
H. Secure the entries in the designated area.

PRELIMINARY ROUND

A. When it is necessary to move models, only judges and official personnel should handle the models. Extreme care should be taken to avoid damage to the entries.
B. Judges independently assess the entries based on the following:
   1. Judges review and score the Model and Display criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
   2. Judges score the Documentation Portfolio criteria of those top twenty-four contestants to determine the top twelve (12) semifinalist teams.
C. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
   1. To deduct 20% of the total possible points or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

D. Judges determine the twelve (12) semifinalists.
E. Create a sign-up sheet for the semifinal interviews.

SEMIFINAL ROUND

A. Semifinalists report at the time and place stated in the conference program to sign up for an interview time.
B. Manage completion of the on-site interviews.
C. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.
D. At the designated time, return models, displays, and portfolios to student owners after verifying official conference identification.